

Sarseej Shrestha

AI/ML Developer · Computer Science Student

Applied AI/ML, medical imaging, uncertainty, data systems, and interpretable model behavior. Building software with careful evaluation and human judgment in the loop.

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EDUCATION

B.S. Computer Science
Southeastern Louisiana University
Expected May 2027

EXPERIENCE

Software Developer
June 2026 – Present
Southeastern Louisiana University · English Department

Modernizing the Early Ruskin Manuscripts archive: TEI-XML publishing in PHP/MySQL, a Dockerized TLS-secured multi-node Elasticsearch cluster, and historical visualizations with Leaflet.js and TimelineJS.

Mathematics Tutor / Senior Mathematics Tutor
September 2023 – May 2026
Southeastern Louisiana University

Taught Statistics, Calculus I–III, Linear Algebra, Differential Equations, and Numerical Methods. Advanced to Senior Tutor, mentored incoming tutors, and developed study guides used across course sections.

AI/ML Intern
June – August 2025
Nawaratna Life School

Built a retrieval-augmented chatbot with FAISS and Flask, and a teacher-facing AI-writing triage classifier. Benchmarked precision and recall, with privacy safeguards and human review.

SQL & Data Systems Intern
August 2020 – May 2022
Nawaratna EBS

Moved paper records into a relational SQL database. Built SQL reports and Python dashboards, used OpenCV for document processing, and analyzed remote-learning survey feedback.

PROJECTS

OncoLA

July – August 2026

Turning a patient's everyday text message into information a care team can use. Semifinalist, Ochsner Health / ASCO Healthcare Hackathon. Draft SOAP notes require clinician approval.

LIDC Reader Disagreement

August 2026

When radiologists disagree, can a model's uncertainty help identify the difficult cases? Predeclared analyses, blinded review, and patient-level split validation keep the investigation accountable.

Lung Nodule Classification

February – July 2026

A three-branch model that makes room for uncertainty alongside a lung-nodule classification. A dedicated uncertainty-estimation branch complements a three-branch ResNet-18 architecture.

RIFT-NYC

July 2025 – February 2026

Bringing physical constraints into models of pipe stress and mechanical systems. PDE-based and variational methods, developed through cross-functional collaboration.

TECHNICAL SKILLS

Application & data systems

Next.js, TypeScript, React, Prisma, SQLite, SQL, PHP, MySQL, APIs

Machine learning & scientific computing

Python, ResNet-18, uncertainty estimation, RAG, FAISS, Flask, OpenCV, TensorBoard, PDE-based and variational methods

Infrastructure & research tools

Docker, Elasticsearch, TLS, automated testing, TEI-XML, Leaflet.js, TimelineJS

Sarseej Shrestha · Curriculum vitae